Estimating Individual Building Heights by Integrating Spaceborne LiDAR and Multisource Remote Sensing Data: A CNN–Transformer Model and a Semi-Supervised Sample Augmentation Approach

Shouhang Du, Hao Liu, Jianghe Xing, Xiuyuan Zhang, Jieyi Zhang, Xunyu Guan, and Shihong Du

***Abstract*—Building height is a key metric in transitioning urban analysis from 2-D to 3-D perspectives. It serves as a fundamental indicator for assessing urban development, population density, and energy consumption. Regardless of whether machine learning or deep learning methods are used, accurate height estimation inevitably relies on large quantities of labeled building height samples for model training. However, the existing reference datasets often suffer from ambiguity due to floor-to-height conversions and temporal obsolescence. The advent of spaceborne light detection and ranging (LiDAR) provides a promising avenue to overcome these limitations, enabling the acquisition of large-scale, high-precision, and time-sensitive reference height data. Moreover, while buildings are inherently individual and discrete units, most existing height estimation studies operate at gridded scales, thereby obscuring interbuilding height variability. In this study, we propose a novel approach to estimate individual building heights by integrating spaceborne LiDAR with multisource remote sensing data. First, leveraging building footprint data, we derive reference height samples by jointly retrieving building heights from the Ice, Cloud, and land Elevation Satellite-2 (ICESat-2) and the Global Ecosystem Dynamics Investigation (GEDI). Then, we construct a multimodal feature set by extracting seasonal features from a combination of multisource datasets, such as Sentinel-1, Sentinel-2, and SDGSAT-1. To model the spatially complex height distribution, we propose the neighborhood-aware convolutional neural network (CNN)–transformer network (NeCT-Net), which estimates building heights at the pixel level. To address the scarcity of tall building samples, we incorporate a semi-supervised learning paradigm. Specifically, pseudo-labels for high-rise buildings are generated using predictions from a teacher model and iteratively refined through self-training, thereby enhancing the student model's learning capacity in tall-structure scenarios. Finally, we propose a postprocessing approach that constrains height predictions using building morphological information, enabling the conversion from pixel-level predictions to vector-based building-level estimations. We apply the proposed method to two iconic urban areas—Beijing and Seattle. The results demonstrate strong predictive performance, with root mean square error (RMSE), mean absolute error (MAE), and R values of 9.01 m, 6.08 m, and 0.93 for Beijing and 8.72 m, 5.31 m, and 0.79 for Seattle. Compared to baseline methods, our approach reduces RMSE by 29.66% in Beijing and 7.43% in Seattle, while improving R by 47.62% and 46.06%. When benchmarked against the existing individual building height products, our model achieves RMSE reductions of 31.43% and 21.16% and R improvements of 57.63% and 23.44% in the two cities. These results highlight the methodological advancement and robustness of our approach. This work presents a new paradigm for large-scale, individual building height estimation through the integration of spaceborne LiDAR and multisource remote sensing data. The source code will be made available at https://github.com/Lunar-Elf/NeCT-Net**

***Index Terms*—Convolutional neural network (CNN)–transformer, Global Ecosystem Dynamics Investigation (GEDI), Ice, Cloud, land Elevation Satellite-2 (ICESat-2), individual building height estimation, semi-supervised learning.**

I. Introduction

As VITAL carriers of human life, cities serve as central hubs for economic, cultural, and technological development [1], [2]. With the advancement of urbanization and the continuous improvement in living standards, cities are playing an increasingly important role in driving economic growth, fostering innovation, providing employment opportunities, and improving quality of life [3]. Buildings, as key components of the physical space of cities, are the carriers of social activities and daily life. Among them, building height is one of the critical indicators for characterizing urban spatial structure, directly affecting urban skyline, population density, land use efficiency, and urban climate, among other aspects [4]. Therefore, rapidly and accurately estimating large-scale building heights in cities holds significant practical and theoretical value for urban planning, land resource management,

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TABLE I
TYPICAL PUBLICLY AVAILABLE LARGE-SCALE BUILDING HEIGHT DATASETS

Code	Dataset	Data Source	Time	Resolution	Coverage	Reference	Method
1	BHG [12]	Sentinel-1/2	2015	10 m	Germany	3DBM	SVM
2	CNBH-10 m [13]	Sentinel-1/2, PALSAR, LUOJIA-01	2020	10 m	China	Amap	Random Forest
3	EUC [14]	Landsat, Sentinel-1	2015	1 km	Europe, USA, China	Amap	Random Forest
4	3D-GloBFP [15]	Sentinel-1/2, PALSAR, Nighttime light	2020	Vector	Global	ONEGEO, Microsoft, Baidu Maps, EMU	XGBoost
5	Northern Hemisphere [16]	Sentinel-1/2, Google Earth Satellite	2020	2.5 m	Europe, USA, China	Microsoft, Amap, EUBUCCO	Efficient-Net
6	GABLE [17]	Beijing-3	2023	Vector	China	DFC 2023	SFFDE

environmental impact assessment, and smart city development [5], [6], [7].

Remote sensing technology is a commonly used means for estimating building height, as it enables the acquisition of large-scale ground information in a short period. In recent years, numerous studies have emerged in the field of building height estimation. High-resolution and high-accuracy building height information can be obtained using light detection and ranging (LiDAR) point clouds, multiview stereo image pairs, and high-resolution satellite imagery. Sun and Salvaggio [8] achieved fast and fully automated 3-D building detection and modeling using airborne LiDAR point clouds. Chen et al. [9] developed a building height estimation method by integrating photogrammetry and deep learning to extract high-resolution building footprints and estimate building height using GaoFen-7 stereo imagery. Zhao et al. [10] extracted building height in parts of Shanghai by integrating spaceborne LiDAR and high-resolution Google Earth satellite imagery. However, these methods face challenges, such as high data acquisition costs, complex data processing, and limited data coverage [11], and cannot meet the requirements of urban planning and resource management applications for large-scale, time-series, and rapidly acquired building height data.

More recent progress in the field lies in collaborative approaches using multisource remote sensing data, such as multispectral and synthetic aperture radar (SAR), which can achieve building height estimation at national, continental, and even global scales. Representative publicly available large-scale building height datasets are shown in Table I [12], [13], [14], [15], [16], [17]. At the national scale, Frantz et al. [12] combined Sentinel-1 SAR and Sentinel-2 time series to estimate building heights in Germany at 10-m resolution using the support vector machine (SVM) model. At the continental scale, Li et al. [14] mapped the 3-D urban structure of Europe, the United States, and China at 1-km resolution using MODIS, Landsat, Sentinel-1 data, and a random forest model. At the global scale, Che et al. [15] estimated building heights worldwide using an XGBoost model based on multisource remote sensing data. Currently, large-scale building height estimation mainly relies on three processes: 1) collection or creation of building height samples; 2) feature extraction for building height estimation; and 3) height prediction using machine learning or deep learning.

In the studies listed in Table I, the collection of a large number of reference data samples for training machine learning or deep learning models is inevitable for building height prediction. Especially in building height estimation using machine or deep learning, the quality of the sample dataset directly affects the model's generalization capability and prediction accuracy [18]. Current studies mainly rely on three types of reference data: statistical data based on building floor counts, height data interpolated from digital terrain models (DTMs), and spaceborne LiDAR data. First, when the study area is within China, researchers often use building floor count data obtained from Amap or Baidu Maps and then convert it to height assuming 3 m per floor [19]. This method has significant drawbacks: the data update cycle is long, with the existing floor data collected in 2017 or 2019, leading to temporal mismatches and data gaps. Moreover, when converting floor counts to height, most studies do not consider functional differences among buildings—for example, commercial buildings often have floor heights of 4–5 m. Second, due to the advancement of airborne LiDAR technology and the increasing availability of open-access datasets, high-precision height data are often derived internationally from interpolated DTMs generated by airborne LiDAR point clouds or high-resolution aerial photogrammetry. The most widely used and extensive dataset is from Microsoft, released in 2017, which provides height data derived from aerial photogrammetric DTM differences [16]. However, due to the high cost of data acquisition, the coverage of such data is limited and rarely updated. Apart from cost, data timeliness is also a concern: with continuous changes in urban structures,

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outdated data may no longer be suitable for future research. Third, with the availability of spaceborne LiDAR data, the challenge of acquiring high-accuracy, large-scale, and timely sample data has been alleviated [20]. LiDAR measures the time it takes for a laser pulse to return after hitting the surface, providing elevation information for the ground and vegetation. Examples include Global Ecosystem Dynamics Investigation (GEDI) and Ice, Cloud, and land Elevation Satellite-2 (ICESat-2). GEDI employs advanced LiDAR to measure the distance from the International Space Station to Earth's surface. ICESat-2 is the first satellite carrying the Advanced Topographic Laser Altimeter System (ATLAS) LiDAR, capable of directly obtaining elevation profiles along its orbit over urban areas [21]. Recently, GEDI and ICESat-2 have been applied in urban building height estimation, but there remain issues. First, few studies integrate them. Each single data source has limitations. For ICESat-2, excessive noise photons make it challenging to distinguish building and ground photons. For GEDI, the large footprint diameter makes it difficult to isolate photons representing the target building. Second, the sample distribution is skewed. Due to the beam-scanning nature of LiDAR data, it forms a striped coverage, and because tall buildings tend to cluster, they may be missed, resulting in an imbalanced sample distribution, which in turn limits the model's ability to learn features of tall buildings. Overall, current building height estimation studies, whether based on machine learning or deep learning, rely on supervised learning. However, due to the lack of tall building samples and outdated reference data, most models underperform for tall building prediction, and some studies only provide results for low-rise and mid buildings [13]. Semi-supervised and weakly supervised learning methods can use easily obtainable labels, reducing manual labeling costs and dependence on large labeled datasets [22]. Therefore, introducing semi-supervised learning can effectively address the issue of lacking tall building samples.

The use of mid- and high-resolution multisource Earth observation data to compute index features for rapid and large-scale building height estimation has become feasible. Tang et al. [23] calculated 20 spatial variables from Sentinel-1/2 and digital surface model (DSM) data and used a random forest model to estimate building heights on 300-m grids in New York and Shenzhen. Furthermore, Wu et al. [13] combined Sentinel-1/2, digital elevation model (DEM), and World Population data, computed various building and shadow-related indices, such as normalized difference built-up index (NDBI) and combinational shadow index (CSI) [24], and used a random forest model to estimate building heights at 10-m resolution across China. Numerous studies demonstrate that machine learning methods integrating Sentinel, DEM, and other multisource data have become mainstream for large-scale urban 3-D structure extraction. However, these methods have common limitations. Most studies only consider optical and SAR index features from satellite imagery, ignoring social and economic factors closely related to building height. In addition, the features extracted from multisource data vary greatly in scale, ranging from 10 m to 1 km, and using the features of different scales introduces uncertainty.

Current building height estimation methods based on multisource remote sensing data still mostly rely on feature-dependent machine learning models. As shown in products 1–4 in Table I, random forest, SVM, and similar models are used to construct regression relationships between reference data and remote sensing features. Furthermore, Ma et al. [21] introduced GEDI-derived height samples and used a random forest model to map building heights in the Yangtze River Delta. These studies confirm the feasibility of using machine learning with multisource remote sensing data to generate global building height maps. However, machine learning methods still exhibit several limitations. First, their ability to extract spatial features remains inadequate, resulting in prediction accuracy that heavily depends on manually computed indices and showing limited adaptability to data heterogeneity. Moreover, the existing machine learning-based approaches commonly estimate building heights on a per-pixel or gridded basis, where each pixel's predicted height depends solely on its own feature set, without incorporating contextual information from neighboring pixels. As a result, the prediction accuracy is difficult to ensure, often constrained to mid- and low-rise buildings and typically presented only at the grid scale.

Deep learning models can autonomously extract and exploit spatial features, model spatial structures in imagery, and offer strong generalization capabilities [25], [26]. In recent years, deep learning has been increasingly applied to building geometry parameter modeling. Yang et al. [20] designed the T-SwinUnet with a temporal attention module, using Sentinel-1 SAR and Sentinel-2 multispectral time-series data to extract 10-m building footprints and heights in the regions, such as the Netherlands and Switzerland. Furthermore, Sun et al. [17] used Beijing-3 imagery and designed a parallel network to simultaneously estimate building height and extract rooftops with fine-grained classification. These studies show that when deep learning is introduced, it is often aimed at obtaining building footprints during height estimation. However, building footprints vary depending on the data source and mapping representation, whereas building height should remain a fixed value across datasets. Thus, this study focuses on estimating building height. Existing deep learning studies still face the following issues. First, the patch-based methods, although sometimes adopted, are prone to boundary artifacts and seam effects near object edges, and they often involve large parameter spaces and heavy computational demands; second, most networks are designed with separate branches for footprint extraction and height estimation and cannot handle sequential data; third, the estimated height results are still in raster pixel form, lacking studies on individual building height estimation.

In summary, current studies have the following problems.

1) Most reference data are derived from floor counts multiplied by 3 m and are outdated. Spaceborne LiDAR retrievals of tall buildings are insufficient for training models to learn tall-building features.
2) When using multisource data, current studies often have limited feature sources and do not adequately consider socioeconomic attributes. The resolutions of

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data sources vary widely (10 m–1 km), and combining features of such differing scales introduces uncertainty.

3) Most current methods use machine learning models with poor spatial feature extraction capabilities, leading to limited accuracy. Deep learning-based methods produce boundary seam artifacts, fail to fully incorporate neighborhood pixel information, and estimate height at the pixel or grid level, obscuring the height attributes of individual buildings.

To address these issues, this study proposes a semi-supervised learning-based neighborhood-aware convolutional neural network (CNN)–transformer network (NeCT-Net) model to estimate large-scale building heights by integrating spaceborne LiDAR and multisource remote sensing data. Building height samples are retrieved from both ICESat-2 and GEDI to construct the ICESat-2-GEDI Building Height Reference (IG-BHR) dataset. Sentinel-2 multispectral, Sentinel-1 and PALSAR polarization, SDGSAT-1 nightlight, NASADEM, and point of interest (POI) data are selected to describe building height information from various dimensions and construct feature models. Neighborhood information is fully considered by using the features of the central pixel and its surrounding pixels as input units to model the relationship with the center pixel's height. While using CNNs to efficiently extract local features, the transformer module is introduced to enhance spatiotemporal dependency modeling and capture long-range dependencies within the input feature dimension for height estimation. The idea of self-training in semi-supervised learning is adopted by using predicted tall building samples as pseudo-labels to retrain the model and improve tall-building prediction capabilities. A method for estimating individual building heights constrained by building morphology is proposed to map pixel-level results to vector building units, ultimately achieving accurate estimation of individual building heights.

The main contributions of this study are as follows.

1) A method for generating reference building height samples by combining ICESat-2 and GEDI data is proposed. This is the first time semi-supervised learning has been introduced to building height estimation. The predicted results from the first round are used to create pseudo-labels for high-rise buildings, addressing the issue of insufficient high-rise samples.
2) A multitemporal feature model combining optical, SAR, terrain, economic, and social data is proposed. By integrating multisource remote sensing data from Sentinel-1/2, PALSAR, SDGSAT-1, POI, and NASADEM, the height characteristics of buildings are comprehensively described from multiple perspectives.
3) An NeCT-Net is proposed to deeply explore the relationship between multisource heterogeneous features and building height. A building height estimation method constrained by building morphology information is also introduced, which maps the height from raster pixel units to building vectors, achieving high-precision estimation of individual building heights.

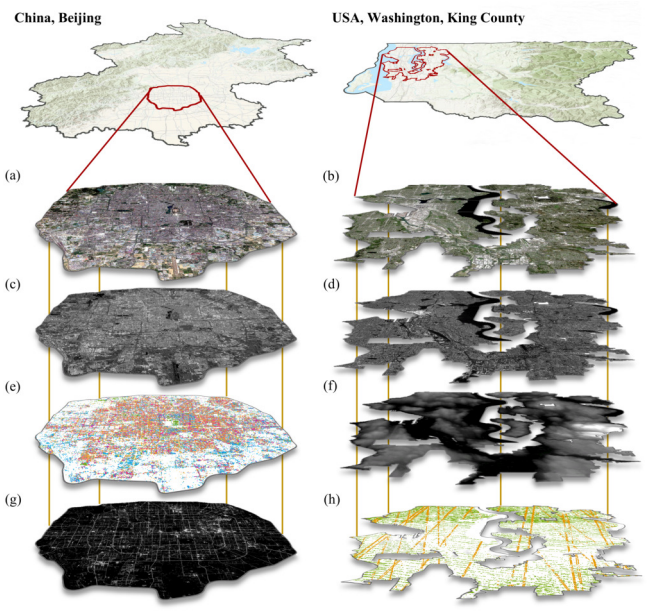


Fig. 1. Overview of the study area and data sources. (a) Sentinel-2 in Beijing. (b) Sentinel-2 in Seattle. (c) Sentinel-1 in VV. (d) PALSAR HH. (e) POI. (f) DEM. (g) SDGSAT-1. (h) Spaceborne LiDAR.

II. Study Area and Data

A. Study Area

This study selects two famous cities, Beijing and Seattle, to evaluate the proposed method for estimating large-scale building heights, as shown in Fig. 1. The area within Beijing's Fifth Ring Road is the core region of China's capital, covering an area of 668 km^2, and includes diverse architectural forms ranging from low-rise alleys to modern skyscrapers. Seattle is located in King County, WA, USA, and the study area includes not only Seattle, the county seat, but also several cities within the bay area, such as Burien and Bellevue, with a total area of approximately 583 km^2. The Seattle metropolitan area has both highly developed urban centers and numerous low-density residential and suburban areas, providing a distinct architectural layout and height distribution pattern compared to Chinese cities. The significant differences in building height distribution between these two areas offer a rich set of building samples for the study, helping to assess the model's performance in handling diverse architectural patterns. It also provides an opportunity to verify the applicability and accuracy of the estimation method in different urban environments.

B. Data

1) Spaceborne LiDAR Data: ICESat-2, a satellite mission initiated by NASA in September 2018, is equipped with the ATLAS as the main tool for measuring the Earth's surface, including ice sheets, glaciers, sea ice, and vegetation canopies. ATLAS emits laser pulses at intervals of 3.3 km along its orbital path to measure surface heights. Each pulse pair is divided into a strong beam and a weak beam with a 4:1 energy ratio, with a 90-m separation between beams. Previous studies have confirmed that building heights can be retrieved from

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TABLE II
FEATURE DATA SOURCES AND DESCRIPTIONS

Category	Datasets	Acquisition time	Resolution	Data Source
Optical	Sentinel-2	2022	10/20 m	"COPERNICUS/S2_SR" (Collection Snippet in GEE)
SAR	Sentinel-1	2022	10 m	"COPERNICUS/S1_GRD"(Collection Snippet in GEE)
	PALSAR	2022	25 m	"JAXA/ALOS/PALSAR/YEARLY/SAR_EPOCH"(Collection Snippet in GEE)
Economical	SDGSAT-1	2022	10 m	https://www.sdgsat.ac.cn/
Social	POI	2020	Vector	https://ditu.amap.com/
Terrain	NASADEM	2000	30 m	"NASA/NASADEM_HGT/001"(Collection Snippet in GEE)

the ICESat-2 L2A Global Geolocation Photon Data [23]. To expand the spatial coverage of ICESat-2 data, we used all available data from the L2A product ATL03 from both day and night, spanning from January 2021 to December 2023.

In addition to ICESat-2, the GEDI LiDAR waveform system, installed on the International Space Station, was used. GEDI provides valuable information about land surfaces and has been used in several studies for estimating vegetation canopy heights [27]. In recent years, it has also been employed for building height estimation [11]. We used GEDI L2A V2 data collected from January 2021 to March 2023 to retrieve building heights. According to prior evaluations, the horizontal geolocation error of GEDI Version 2 data is approximately 10.3 m, while that of ICESat-2 ranges from 2.5 to 4.4 m. Both fall within the 10-m scale [28], [29], which corresponds roughly to the pixel size of the multisource feature sets constructed in this study using Sentinel-1/2, PALSAR, and other 10-m resolution imagery. As such, the horizontal positioning errors are approximately equivalent to a single pixel and have limited impact on the displacement of sample height values. Since the two satellites follow different patterns and access the same region at different times, combining both spaceborne LiDAR systems helps to expand the coverage and mitigate the system error impacts of individual data sources. To ensure both the quality and spatial coverage of reference data, the acquisition period for the spaceborne LiDAR data was set to span from 2021 to 2023. Given that the construction of buildings is a prolonged process—typically requiring 1–3 years from groundbreaking to completion depending on building type—we consider the retrieved heights within this time window to be representative of the corresponding building heights.

2) Input Features for Height Estimation: The feature data sources used for building height estimation are shown in Table II, and all data preprocessing was performed on the Google Earth Engine (GEE) platform. The data are categorized into seasonal data—primarily Sentinel imagery—and nonseasonal data, including PALSAR polarization data, NASADEM terrain data, SDGSAT-1 nighttime light imagery, and POI data. The feasibility of using seasonal composite imagery for large-scale building height estimation has been validated by Lyu et al. [30] and Frantz et al. [12]. As the solar elevation angle at local noon gradually increases from the winter solstice to the summer solstice, the maximum shadow length cast by buildings correspondingly decreases. Since Sentinel satellites operate on sun-synchronous orbits with fixed overpass times, seasonal variations directly influence the shadow lengths captured in the imagery [12]. Moreover, seasonal features enrich the expression of land surface characteristics and maximize the extraction of diverse contrasts between buildings and their surrounding environments.

Sentinel-2 A and B satellites provide globally accessible multispectral data. The study uses four bands (bands 2, 3, 4, and 8) with 10-m resolution and six bands (bands 5, 6, 7, 8A, 11, and 12) with 20-m resolution, all radiometrically calibrated and atmospherically corrected in GEE. Images from January, April, July, and October of 2022, filtered for cloud cover, were processed using the QA60 band of Sentinel-2 for cloud removal. The median composition algorithm on GEE was then applied to generate seasonal composite images, filling the gaps in the cloud-removed images. This process resulted in four high-quality seasonal composite images. Similarly, Sentinel-1 data were extracted quarterly by taking the median of the VV and VH polarization backscatter intensities [31].

To expand the feature set and provide auxiliary information for building height prediction, nonseasonal datasets were also used. PALSAR, widely used for estimating vegetation vertical structures, has also been employed to complement building microwave dimensional information in the existing building height estimation studies. We used the 25-m global PALSAR/PALSAR-2 annual mosaic data from 2022 in GEE, with HH and HV polarization scattering information supplementing the Sentinel-1 polarization data.

The study also incorporated SDGSAT-1's advanced night light data. The Glimmer Imager for Urbanization (GIU) on SDGSAT-1 provides nighttime light data with 10-m resolution, offering varying light intensity levels for urban and rural areas. The data used here are the fourth-level product, which was orthorectified using ground control points and a DEM based on the first-level standard radiometric correction. This process ensured that the data were formatted appropriately and ready for use in the study.

POI data provide functional attributes of buildings, which can effectively improve the accuracy of building height

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estimation [32]. Similar building types tend to follow patterns in horizontal distribution and vertical height, which POI data abstract and quantify. POI data for Beijing were obtained via an API from Amap (https://ditu.amap.com/), while POI data for Seattle were downloaded from OpenStreetMap (https://openmaptiles.org/). In this study, we refer to common building types and the Chinese Urban Land Use and Planning Standard (GB50137-2011) to classify buildings into six categories: residential, commercial, public services, science/education/culture, scenic spots, and transportation facilities. Based on this classification scheme, the original POI data types are reclassified accordingly. These categories often represent distinct building systems in real-world urban environments. The original POI data were then reclassified. Kernel density estimation was applied to convert the POI data into continuous density surfaces that reflect spatial distribution differences, thereby enhancing the descriptive capability of the feature set for buildings.

NASADEM, providing a high-quality DEM with 30-m resolution, was used to account for terrain factors that significantly affect solar altitude angles, which, in turn, cause buildings of similar height to appear differently in satellite imagery due to variations in radiometric values.

3) Building Footprint and Height Data: The building footprint data used in this study are from 3D-GloBFP [15], primarily sourced from open data platforms, such as ONEGEO, Microsoft, and Baidu Maps. The building footprint data are mainly used to assist in retrieving building heights from laser photons and serve as the base data when mapping heights from raster pixel units to building vector heights in the final results. In addition to the validation dataset provided by IG-BHR, we further adopt two commonly used reference datasets from prior studies to cross-validate the reliability of the predicted individual building heights. This multiangle evaluation enhances the robustness of our experimental assessment.

1) For Beijing, auxiliary validation data were obtained from Amap (https://ditu.amap.com/). Amap collected building floor number information across multiple Chinese cities through field surveys and publicly released the data in the vector format. The dataset used in this study was acquired in 2019. The floor number data from Amap served as a reference to evaluate the correlation between the estimated building heights in Beijing and common reference datasets. Following the method of Wu et. [13] and others, floor height is assumed to be 3 m, and the floor count is converted into height. According to Cao and Huang [33], Amap's height dataset shows a strong agreement with actual building heights, with a coefficient of determination of 0.97 and a root mean square error (RMSE) of 1.19 m. It has also been used as reference data in several works that produce building height datasets for China [13].
2) For the Seattle metropolitan area, auxiliary validation data were derived from the building height training dataset released by Microsoft in 2017 (https://wiki.openstreetmap.org/wiki/Microsoft_Building_Footprint_Data#March_2017_Release). This dataset includes approximately 9.8 million building footprint records with high-accuracy height information across 44 U.S. states. The data are publicly available in vector format, with building heights (m) obtained through interpolation of DTM data. These data have been used by Microsoft to predict building heights across the United States and were used as reference data by Cao and Weng [16] to successfully predict building heights in the Northern Hemisphere.

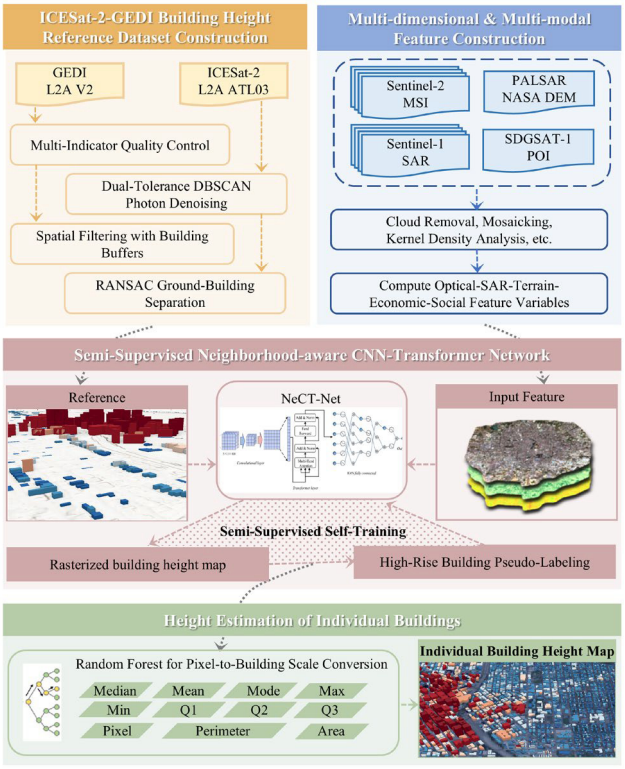


Fig. 2. Method flowchart.

III. Methodology

This study uses spaceborne LiDAR satellite data, combined with multisource remote sensing data, such as Sentinel and SDGSAT-1, to perform large-scale building height estimation. Overall, our approach is divided into four parts, as shown in Fig. 2.

1) We use ICESat-2 and GEDI to produce the IG-BHR dataset, which serves as reference data for training the deep learning model in the third step. This process primarily involves photon denoising, filtering, classification for ICESat-2, and multi-index quality screening for GEDI, as well as using building contour data as constraints to map relative photon height to building height.
2) We calculate multiseasonal features using multisource remote sensing data, such as Sentinel-1, Sentinel-2, and SDGSAT-1, and construct a multimodal feature set, which serves as input features for the third step deep learning model.
3) We propose the NeCT-Net for building height estimation in the study areas. By introducing the concept of

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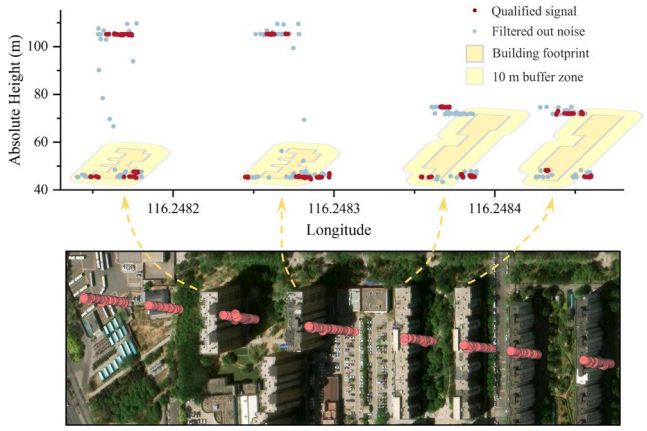


Fig. 3. ICESat-2 photon processing effect.

semi-supervised learning, we create high-rise pseudo-labels based on initial predictions and enhance the model's ability to learn the features of high-rise buildings through self-training.

4) We propose a method for estimating individual building height using building morphology as a constraint, mapping height from raster pixel units to building vector data, and ultimately estimating individual building heights.

A. IG-BHR Dataset Construction

ICESat-2 ATL03 products provide the latitude, longitude, and elevation of each photon, allowing identification of the height of building tops and surrounding ground photons in a local area, calculating the relative height of buildings. Due to the large data volume from ICESat-2's long orbit, we introduce building contour data to assist, creating a 10-m buffer around the building rooftops to retrieve photon heights for each individual building [34]. The raw data contain a lot of background noise photons. We first use the DBSCAN algorithm with a tolerance of 2.1 m to identify and remove about 90% of the scattered noise [35]. Then, using a tolerance of 1.4 m for DBSCAN to further remove smaller clusters that deviate from the main target. RANSAC linear fitting [23] is applied to eliminate points with vertical distances exceeding three standard deviations of the outliers, retaining the ground and rooftop clusters with small slopes and significant elevation differences. The final processed photon data are shown in Fig. 3. For each building, we compute the difference between the maximum height (rooftop cluster) and the minimum height (ground cluster) as a candidate height. Irregular values (with height differences less than 2.5 m) are further filtered out to obtain the relative height of the building.

Unlike ICESat-2, GEDI L2A V2 data directly provide relative height indices derived from geo-located waveforms. In previous studies, these relative height indices were considered to represent the relative height of corresponding buildings. Based on the existing research, this study uses the relative height index, rh95, as the reference building height [21]. High-quality data are selected by setting the quality flag (quality_flag = 1) and degradation flag (degrade_flag = 0), where the quality flag of 1 meets waveform energy, sensitivity, amplitude, and real-time surface tracking quality standards, and the degradation flag of 0 avoids orbital attenuation effects. Additionally, we exclude invalid data with fewer than two detected modes (num_detectedmodes < 2) to ensure both ground and building double-echo features are included and exclude observations with a sensitivity of less than 0.9 (sensitivity < 0.9) [11]. The filtered GEDI photon points generate a 2-m buffer, and the height attribute is mapped to building vectors via spatial join.

By incorporating building vector data as an auxiliary, the relative height previously recorded by photon points is mapped to buildings, helping to mitigate the positional errors inherent in spaceborne LiDAR data. The building height retrieved by both datasets is recorded in the vector feature fields, and the data are merged to form the IG-BHR. The centroid of the vector polygons is then calculated, and the retrieved building height result is assigned to these centroid points, turning them into sample points with height attributes. 70% of these points are used for training the deep learning model, and the remaining 30% is used for validation.

B. Multidimensional and Multimodal Feature Construction

The multidimensional features include five parts: optical, SAR, terrain, economic, and social, with the variables listed in Table III.

For optical features, we use Sentinel-2 data and select 21 features, including ten band features, three vegetation indices, and eight indices related to vegetation, water bodies, buildings, and shadows, such as normalized difference vegetation index (NDVI), enhanced vegetation index (EVI), normalized difference water index (NDWI), land surface water index (LSWI), NDBI, normalized saturation-value shadow index (NSVDI), shadow-enhanced index (SEI), and CSI. These features are synthesized across four seasons, forming a set of 84 optical variables. Among them, vegetation indices assist in distinguishing shadows caused by tall vegetation and correcting scattering interference recorded in SAR data [14]; water-related indices help mitigate height estimation errors arising from misclassifying water bodies as false shadows; NDBI facilitates the identification of built-up areas while suppressing nonurban regions; and shadow indices enhance the model's capacity to recognize shadow information, contributing to a more effective learning of the relationship between building shadows and height.

For SAR features, we use Sentinel-1 data and select three variables: VV and VH polarization, as well as the VVH feature containing dual-polarization information [36]. These features are also synthesized across four seasons. Combined with PALSAR's HH and HV polarization features, a total of 14 SAR features are formed. These polarization features, measured from different scattering perspectives, capture the microwave radiation scattered by buildings and are indicative of vertical structural geometry as well as electromagnetic material properties.

For terrain features, we use NASADEM data, along with slope and aspect features, creating a set of three terrain features. Terrain characteristics—such as slope and

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TABLE III
VARIABLES INCLUDED IN THE FEATURE MODEL

Code	Data Source	Variable	Resolution
1-21	Sentinel-2	Blue; Green; Red; NIR; Red Edge 1–4; SWIR1-2; NDVI; EVI; LSWI; NDWI; NDBI; SEI; CSI; NSVDI; Brightness; Greenness; Wetness	10 m/20 m
22-24	Sentinel-1	VV; VH; VVH	10 m
25-26	PALSAR	HH; HV	25 m
27-29	NASADEM	DEM; Slope; Aspect	30 m
30	SDGSAT	Nighttime light	10 m
31-36	POI	Accommodation; Business; Public Facilities; Education; Scenery; Transportation	Vector

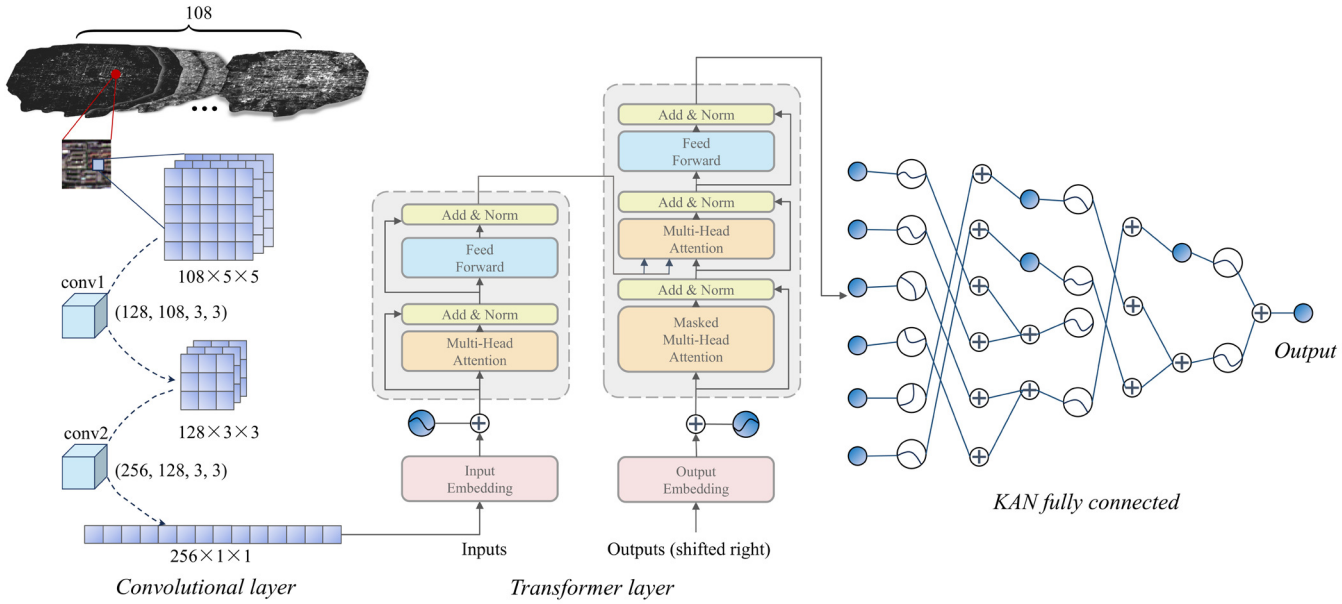


Fig. 4. Network architecture diagram.

orientation—affect local solar elevation angles and, in turn, influence spectral and shadow properties relevant to height estimation.

For economic features, SDGSAT-1's nighttime light imagery is used as a variable, forming a set of 1 economic feature. Building height is partially reflective of local economic development, where higher levels of economic activity are generally associated with taller structures [15].

For social features, we use POI data from six common functional types: residential, commercial, public, cultural, scenic, and transportation, to generate kernel density images, forming a set of six social features. Functional distinctions among buildings are often associated with varying floor heights. Buildings of the same function tend to exhibit regular patterns not only in horizontal spatial distribution but also in vertical height characteristics. Therefore, functional information is an essential component in comprehensively modeling building height.

The data for all features are acquired and preprocessed on the GEE platform. Each variable is resampled to a 10-m resolution using the nearest neighbor method and linearly normalized to the range of 0–1, forming a comprehensive multidimensional feature set that describes building height.

C. Semi-Supervised NeCT-Net

The multimodal feature set, which includes optical, SAR, terrain, economic, and social features, provides information about various aspects of buildings. However, these heterogeneous features increase the demands on the model's ability to extract features. CNN frameworks can partially extract features from different levels of buildings by changing the convolution kernel size and the number of layers, but their ability to capture global information remains limited [37]. Fully connected layers usually use multilayer perceptrons to learn both linear and nonlinear relationships between features, but their weight matrices are fixed, lacking dynamic adaptability to input features. In remote sensing imagery, buildings manifest as cohesive structures composed of multiple adjacent pixels. Accurately estimating the height of a target pixel within a building requires more than isolated information from that pixel; relying solely on 1-D feature vectors

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neglects the essential spatial context. Traditional methods frequently overlook this neighborhood information by treating pixels independently. Therefore, we design NeCT-Net, whose network structure, shown in Fig. 4, consists of three parts: 1) CNN convolution layers; 2) transformer encoder layers; and 3) Kolmogorov–Arnold Networks (KANs) fully connected prediction layers. This design combines the local feature extraction ability of CNN with the global dependency modeling capability of transformer, processing information from low-level texture features to high-level semantic information, and modeling global dependencies in a step-by-step manner. Furthermore, we innovatively use image blocks as an input. In contrast to the existing studies that either aggregate regional statistics or compute texture metrics [13], [15], we explicitly incorporate neighborhood context by feeding all features within a 5×5 image block centered on the target pixel into the network. This strategy enables the model to effectively learn the relationship between local spatial structures and building height, enhancing estimation accuracy for individual pixels through contextual awareness. These image blocks contain information about buildings and their surrounding environment, allowing the model to not only extract the features of the target building but also integrate the surrounding spatial structure features, providing more comprehensive information for height estimation.

The convolutional layers play a crucial role in extracting spatial features from the input remote sensing data. The input feature dimensions of the model are ($\text{batch}_{\text{size}} = 3$, channels $=108$, $H = 5$, and $W = 5$). Initially, two 3×3 convolutional layers are applied sequentially to progressively expand the feature dimensions. After each convolutional operation, the ReLU() activation function is employed, enhancing the model's ability to capture nonlinear patterns. The first convolutional layer uses an input channel of 108 and an output channel of 128, yielding the feature map after the first convolutional layer

$$f_1 = \text{ReLU}\left(\text{Conv2d}_{(3\times3)}\left(x; W_1, b_1\right)\right). \tag{1}$$

The second convolutional layer uses an input channel of 128 and an output channel of 256, resulting in the feature map after the second convolutional layer

$$f_2 = \text{ReLU}\left(\text{Conv2d}_{(3\times3)}\left(f_1; W_2, b_2\right)\right). \tag{2}$$

The final CNN feature map, denoted as f_2, is flattened to transform it into a feature vector with dimensions of $(\text{batch_size}, 256)$, referred to as f_{conv}. The use of smaller convolutional kernels aids in accurately capturing detailed building features. By stacking two convolutional layers and progressively enhancing the feature dimensions, the model elevates low-level texture information to higher level semantic information while preserving key spatial structural features.

The transformer encoder layer serves to globally integrate the local features extracted by the convolutional layers and further captures long-range dependencies in the high-level representations of the feature map. First, the model applies a learnable linear transformation to the convolutional layer's output to adapt it to the input dimension of the transformer. The mapping process is as follows:

$$f_{\text{embed}} = f_{\text{conv}} \cdot W_{\text{embed}} + b_{\text{embed}} \tag{3}$$

where W_{embed} and b_{embed} are the weight matrix and bias term, respectively.

Subsequently, position encoding PE is injected into the feature representations using a position encoding matrix, allowing the model to preserve spatial structural features while also processing sequential information. The position encoding formula is

$$f_{\text{pos}} = f_{\text{embed}} + \text{PE}. \tag{4}$$

The input feature f_{pos} is then linearly transformed to generate query (Q), key (K), and value (V) matrices, which are used to compute attention weights through a dot-product attention mechanism

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^1}{\sqrt{d}_{\text{model}}}\right)V. \tag{5}$$

To allow the model to learn diverse feature representations, the self-attention layer is expanded into a multihead attention mechanism. Assuming there are h attention heads, the final output is the concatenation of all heads

$$\text{MultiHead}\left(Q, K, V\right) = \text{Concat}\left(\text{head}_1, \ldots, \text{head}_h\right) \cdot W^O \tag{6}$$

where each $\text{head}_i = \text{Attention}(Q_i, K_i, V_i)$ and W^O is the output transformation matrix.

The output of the self-attention layer undergoes layer normalization before being fed into the feedforward neural network, resulting in f_{FFN}. This network consists of two fully connected layers, one for nonlinear activation and the other for output transformation. After each attention layer and feedforward network, residual connections and layer normalization are applied. The final output of the transformer module is

$$f_{\text{trans}} = \text{LayerNorm}\left(f_{\text{att}} + f_{\text{FFN}}\right). \tag{7}$$

In the task of building height estimation, long-range dependencies between features are of particular importance. The environmental background information contained in remote sensing data can significantly influence the prediction of the target building's height. By introducing the transformer encoder layer, the model can model long-range spatial dependencies within the feature map, greatly enhancing the comprehensiveness and accuracy of the feature representation. This self-attention-based design enables the model to not only focus on local building details but also fully utilize surrounding information to improve estimation accuracy.

In height estimation, there exists a complex spatial dependency between buildings and their surrounding environment, which is typically difficult to capture using fixed fully connected layer weights. Thus, compared to traditional multilayer perceptron (MLP), the KAN demonstrates significant advantages in feature modeling and capturing nonlinear relationships. MLPs generally learn both linear and nonlinear relationships between features through a series of fully connected layers, but their weight matrices are fixed and lack dynamic adaptability to input features. In contrast, KAN introduces a spline interpolation mechanism in feature modeling, allowing it to process the complex relationships between input features in a more refined manner. Specifically, the spline weights in KAN provide flexible nonlinear mappings, enabling

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the model to dynamically adjust the output based on the input features and capture subtle variations and nonlinear trends between them.

The KAN network consists of multiple fully connected layers that progressively project high-dimensional features into a low-dimensional output space, thereby achieving precise building height estimation. The network captures the complex relationships between features through two nonlinear transformations, and the output is a 1-D predicted building height. The first layer has 256 input units and 64 output units, and the second layer has 64 input units and one output unit. The specific calculation formula is as follows:

$$f_{\text{out}} = W_{\text{KAN2}} \cdot \text{ReLU}\left(W_{\text{KAN1}} \cdot f_{\text{trans}} + b_{\text{KAN1}}\right) + b_{\text{KAN2}} \quad (8)$$

where f_{out} represents the predicted height of the input sample, W_{KAN1} and b_{KAN1} are the weight matrix and bias term for the first layer, respectively, W_{KAN2} and b_{KAN2} are the weight matrix and bias term for the second layer, respectively, and f_{trans} refers to the output features of the transformer module.

Spaceborne LiDAR employs beam scanning, resulting in strip coverage of the data. However, urban high-rise buildings are densely clustered, so the IG-BHR retrieves relatively few high-rise buildings. The sample distribution is significantly skewed, preventing the model from adequately learning the features of high-rise buildings during training when using reference data. Therefore, this study introduces self-training from semi-supervised learning. IG-BHR is used to train the NeCT-Net model as the teacher model. High-rise samples, about 2000, are selected from the building results obtained by the teacher model to create pseudo-labels. These pseudo-labeled samples are combined with IG-BHR data to train the student model.

D. Height Estimation of Individual Buildings

The height of an individual building is decomposed into multiple pixel-level height values in the 10-m resolution gridded building height map, where each pixel represents a partial mapping of the building's overall height. Therefore, a method is needed to capture the nonlinear relationship between these discrete pixel-level heights and the actual building height. Some studies have estimated individual building heights by combining normalized DSM (nDSM)-derived elevation data with building footprint polygons [38]. Various statistical metrics can be used to represent rooftop height by analyzing all pixel-level height values within the building footprint [39], [40]. Tack et al. [40] further demonstrated that the median is robust to outliers during the aggregation process from pixel-level heights to object-level building height and performs well across multiple evaluation metrics. Therefore, this study proposes a building height estimation method constrained by building morphological information. First, building contour vector data is introduced, and a 10-m buffer zone is created as the unit for calculating individual heights [15]. Next, in addition to statistical data, such as the median, mean, mode, maximum, minimum, Q1, Q2, and Q3 of the corresponding region's grid height map within the buffer zone, the perimeter, area, and number of occupied pixels of each building are also extracted. In total, 11 variables are used as predictors, with the corresponding individual heights from IG-BHR as reference values. Then, this study employs a random forest model for regression modeling. Previous studies often inferred the overall building height using a single metric, which may introduce uncertainties. In contrast, the random forest regression model is capable of effectively handling high-dimensional data and complex nonlinear relationships, while also exhibiting strong resistance to overfitting. To balance training time and computational cost under optimal performance, the number of decision trees in the random forest (n_estimators) is set to 100, with the random seed (random_state) set to 42. Finally, the trained model is applied to all individual buildings, mapping the height from grid pixel units to building vectors, thus enabling the estimation of individual building heights.

E. Accuracy Evaluation

The overall accuracy evaluation is primarily based on 30% of the IG-BHR dataset, which is held out for testing. In Section V, the two auxiliary datasets are used to assess the consistency of our predicted results with common reference data through correlation analysis. Referring to the international standards for tall buildings defined by the Council on Tall Buildings and Urban Habitat (CTBUH), as well as commonly used national standards, such as the General Code for Civil Buildings in China, we categorize buildings into five height intervals: low-rise buildings (heights between 0 and 10 m), mid-rise buildings (heights between 10 and 24 m), mid-high-rise buildings (heights between 24 and 50 m), high-rise buildings (heights between 50 and 100 m), and skyscrapers (heights greater than or equal to 100 m). To comprehensively evaluate the accuracy of the model in building height prediction, the model results are analyzed by interval error according to the above height categories, ensuring that buildings of different height categories are included in the evaluation.

Three evaluation metrics are selected: correlation coefficient (R), RMSE, and mean absolute error (MAE). R is a statistical measure of how well the regression model fits, representing the proportion of variance in the dependent variable that can be explained by the model. RMSE is the sample standard deviation of the residuals between the predicted and true values, indicating the stability of the model's predictions. MAE is the average absolute error between the predicted and true values, reflecting the model's accuracy and how close the predicted values are to the actual values

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(\text{BH}_{\text{estimated},i} - \text{BH}_{\text{reference},i}\right)^2}{n}} \quad (9)$$

$$\text{MAE} = \frac{\sum_{i=1}^{n}\left|\text{BH}_{\text{estimated},i} - \text{BH}_{\text{reference},i}\right|}{n} \quad (10)$$

$$R = \sqrt{1 - \frac{(n-1)\sum_{i=1}^{n}\left(\text{BH}_{\text{estimated},i} - \text{BH}_{\text{reference},i}\right)^2}{(n-2))\sum_{i=1}^{n}\left(\text{BH}_{\text{estimated},i} - \text{BH}_{\text{reference},i}\right)^2}} \quad (11)$$

where n is the number of validation samples, $\text{BH}_{\text{estimated},i}$ is the estimated building height for the i building, and $\text{BH}_{\text{reference},i}$ is the reference building height for the i building.

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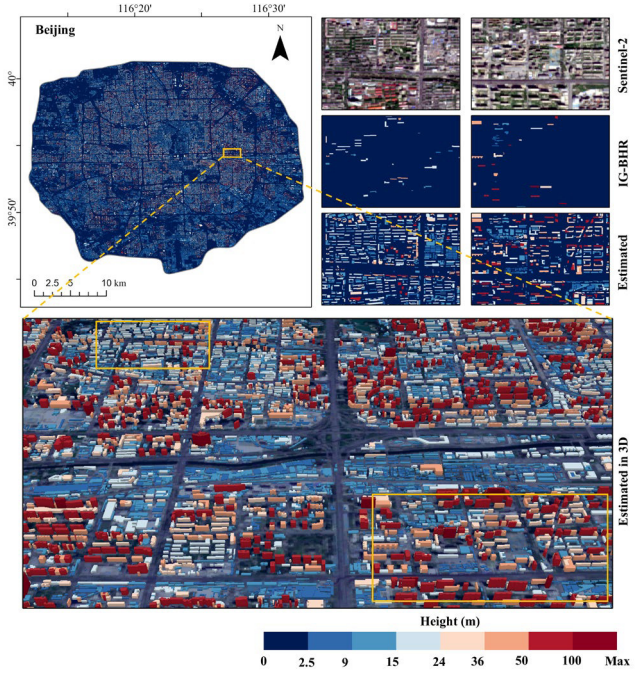


Fig. 5. Estimation results of building height in Beijing.

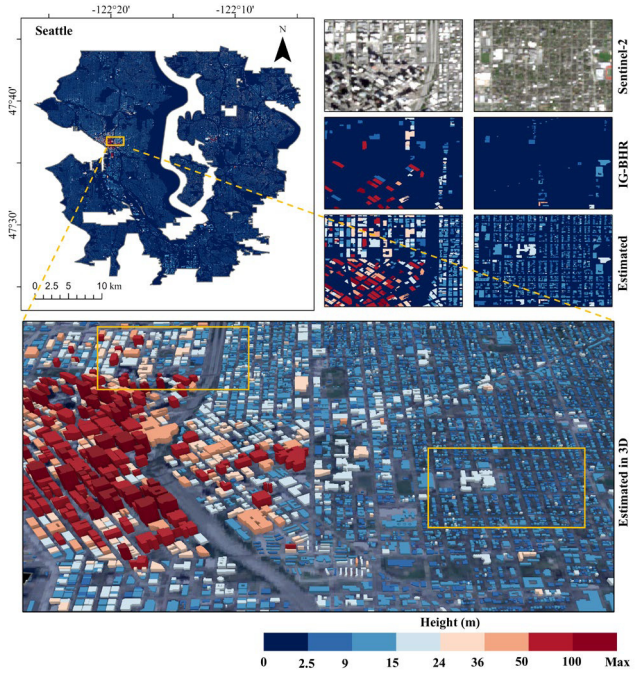


Fig. 6. Estimation results of building height in Seattle.

IV. Results

A. Performance in Two Cities

After self-training, the NeCT-Net model, using multidimensional feature sets from two cities as an input, successfully extracted building height map. Using building shape information as a constraint, the model converted grid pixel units to building vector heights and estimated the heights of individual buildings in Beijing and Seattle. The random forest model constrained by building outlines achieved the scale conversion

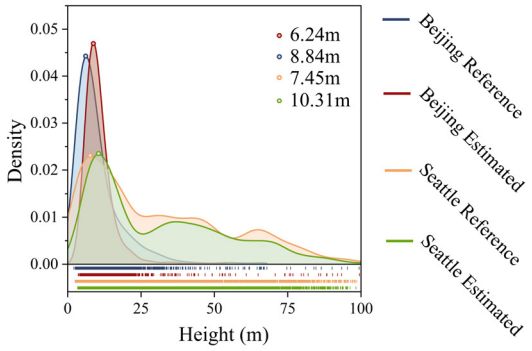


Fig. 7. Probability density curves of building height.

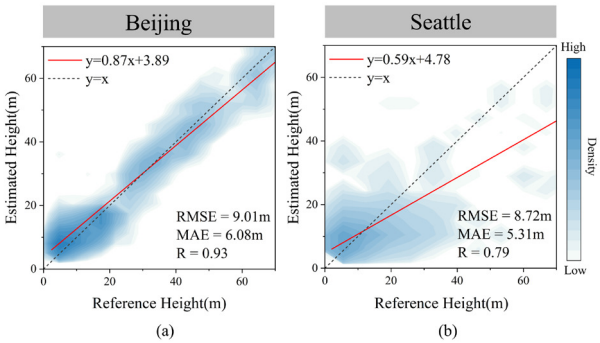


Fig. 8. Accuracy evaluation of the predicted building heights in Beijing and Seattle. (a) Beijing. (b) Seattle.

from pixel to vector units. The building height extraction results for Beijing and Seattle are shown in Figs. 5 and 6, with average building heights of 14.8 and 10.1 m, respectively. By comparing the prediction details of each region with the reference data IG-BHR and the true color composite images of Sentinel-2, it was found that the model accurately identifies buildings of different heights. When stretched according to height for 3-D visualization, the results reflect the variation in building height within the urban area. Benefiting from using a 5×5 feature block as the prediction unit, the model takes into account the influence of surrounding pixels, enabling good prediction accuracy even in areas with mixed high-rise and low-rise buildings, such as in the city's central business district.

The density distributions of the predicted building heights and reference heights in both cities are shown in Fig. 7. The overall trend of the prediction results aligns well with the reference data, with Seattle showing a unimodal pattern and Beijing showing a trimodal pattern. The point density plots comparing predicted building heights with the IG-BHR validation data for the two study areas are shown in Fig. 8. The study confirms that the NeCT-Net-based building height model demonstrates significant generality across cities, maintaining a high correlation and low error. Overall, the prediction results for Beijing are more ideal, with RMSE, MAE, and R values of 9.01 m, 6.08 m, and 0.93, respectively. In Seattle, the accuracy values are 8.72 m, 5.31 m, and 0.79. Beijing has a relatively flat terrain, and the distribution of collected building height samples is rich, so the regression line of the predicted

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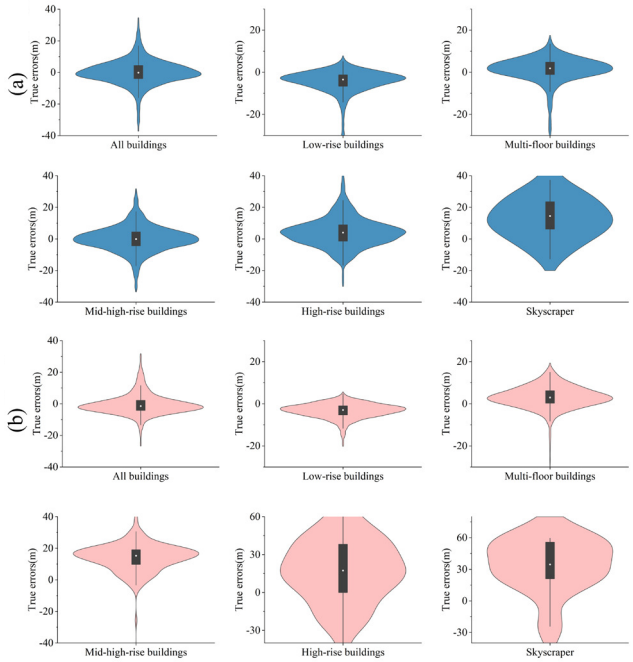


Fig. 9. Violin diagram of the true error in predicted building heights. In each subplot, the width of the shapes represents the relative probability density at the corresponding true error value, with wider shapes indicating a higher quantity of occurrences at that true error value. (a) Beijing. (b) Seattle.

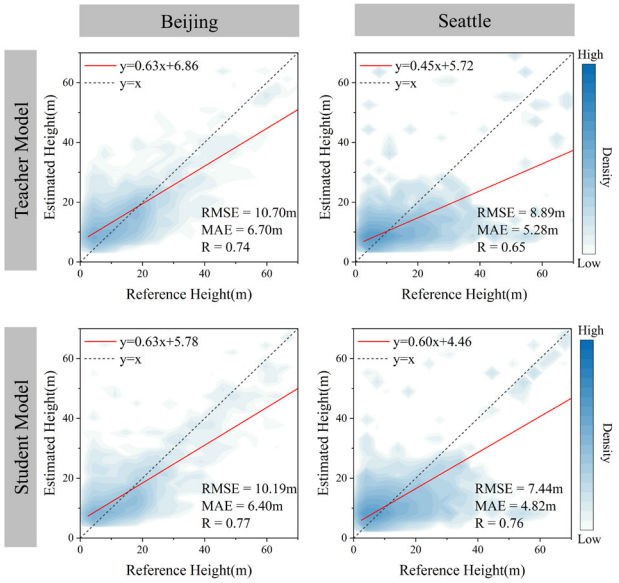


Fig. 10. Accuracy comparison of prediction results between teacher and student models.

values aligns closely with the actual values, with the slope near 1, indicating good linear consistency in height prediction. Seattle, with its more rugged terrain, has a unimodal skewed distribution of building height samples. The fitted line's k-value is slightly lower, and the intercept is relatively higher, with a slightly more dispersed density distribution in the results.

The true errors in the estimated building heights for both Beijing and Seattle were statistically analyzed in five height intervals, and the violin plot of the errors is shown in Fig. 9. On the whole, the error distributions in both cities are centered around 0 m, with the median nearly at zero, indicating that the model neither overestimates nor underestimates building heights. In terms of intervals, the median errors for each height category are similarly close to zero. Specifically, the violin shapes for low-rise buildings, mid-rise buildings, and mid-high-rise buildings are relatively narrow, indicating high prediction stability, with errors controlled within a small range. This shows that for common residential and medium-height office and commercial buildings, the model provides stable and accurate predictions. High-rise and skyscraper buildings have a wider distribution, and the positive long-tail error distribution increases, which is associated with the more complex morphology of these buildings and the relatively few available samples. However, the median errors are still low, and the error proportion to height is also minimal. Overall, despite differences in building styles and sample distributions between Beijing and Seattle, the errors for various height categories in both cities are low and concentrated, with similar error distributions and medians, indicating that the model maintains robust performance and generalizes well across cities. High-rise buildings exhibit greater diversity in functional types, encompassing not only standard residential towers but also small office buildings and mixed-use complexes. The variety of rooftop materials and substantial intraclass variability among samples increase the overall complexity of height prediction for this category.

B. Impact of Semi-Supervised Approach on Estimation Errors for Different Building Heights

The number of high-rise buildings retrieved by spaceborne LiDAR data is low, which is not conducive to the model learning to predict high-rise buildings. Therefore, this study introduces the self-training approach from semi-supervised learning to address this issue. The overall accuracy comparison between the teacher and student models is shown in Fig. 10. After adding high-rise pseudo-labels, the prediction accuracy of the student model significantly improved in all metrics. Specifically, in Beijing, RMSE decreased by 4.77%, MAE decreased by 4.48%, and R increased by 4.05%. In Seattle, RMSE decreased by 16.31%, MAE decreased by 8.71%, and R increased by 16.92%. The slope comparison between the predicted values and actual values also improved in the student model, indicating enhanced linear consistency in the predictions.

To fully demonstrate the impact of this semi-supervised approach on the model's prediction capability, the buildings were divided into five categories according to their heights, and error assessments were performed layer by layer. The true errors of the teacher and student model predictions for both Beijing and Seattle were statistically categorized, and the boxplots of the true errors are shown in Fig. 11. It is evident that after introducing the high-rise pseudo-labels from the teacher model, the student model's predictions for high-rise buildings (100 m > height > 50 m) and skyscrapers (height > 100 m) showed a significant reduction in errors.

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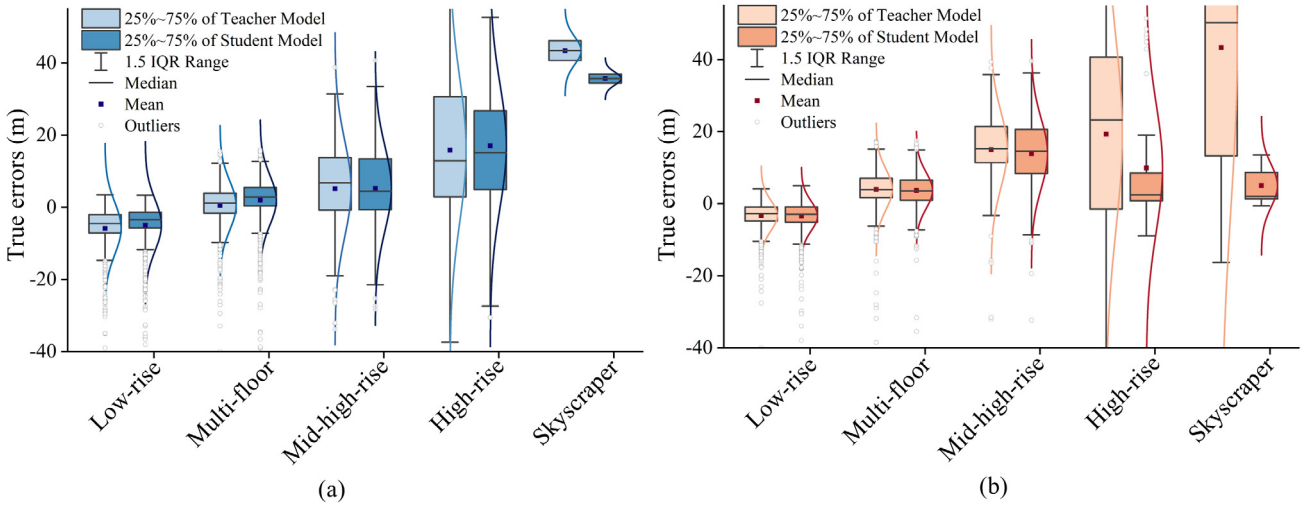


Fig. 11. True error distribution for teacher and student models across different height categories. (a) Beijing. (b) Seattle.

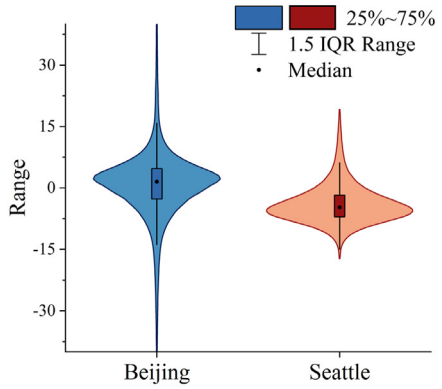


Fig. 12. Violin plots of height differences between our results and commonly used reference datasets.

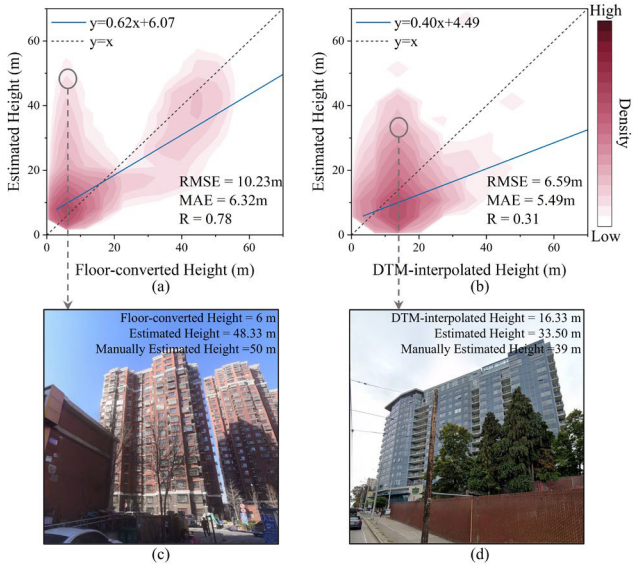


Fig. 13. Comparison between predicted building heights and external benchmark datasets. (a) Scatter density map comparing predicted height with floor-converted height. (b) Scatter density map comparing predicted height with DTM-interpolated height. (c) Street-view image of samples overestimated compared with floor-converted height. (d) Street-view image of samples overestimated compared with DTM-interpolated height.

Specifically, the interquartile range (1.5 IQR range) for the student model in high-rise buildings and skyscrapers noticeably shrunk compared to the teacher model, with the error distribution shifting toward 0. This indicates improved stability, reduced fluctuation in error, and significantly enhanced prediction accuracy. Additionally, for low-rise and mid-rise buildings, the median and mean of the boxplots are close to zero, and the error distribution is more concentrated, indicating that the model's predictions for these building types are quite accurate with small errors.

Although the pseudo-labels in this semi-supervised learning approach contain errors, the results show that after incorporating high-rise building pseudo-labels, the student model's accuracy for low-rise buildings, mid-rise buildings, and mid-high-rise buildings remained largely unaffected. The improvement in accuracy was mainly seen in high-rise buildings and skyscrapers, which demonstrates that the model did not suffer from negative effects due to the inclusion of pseudo-labels in the middle and low-rise height categories. Furthermore, the validation and training sets for both models are independent, with the validation set unaffected by the pseudo-labels. The results suggest that the model did not experience significant error accumulation or overfitting due to pseudo-labels. The experimental results fully demonstrate that the teacher–student training framework effectively improves the model's generalization ability. By learning from the high-rise building pseudo-labels predicted by the teacher model, the student model corrected the biases caused by insufficient samples in the original data, thereby performing better on the same validation set.

V. Discussion

A. Comparison of Building Height Results With Other Common Reference Data

Given that common approaches still rely on floor-count-based building heights and Microsoft's DTM-interpolated heights, we incorporate these two commonly used datasets as

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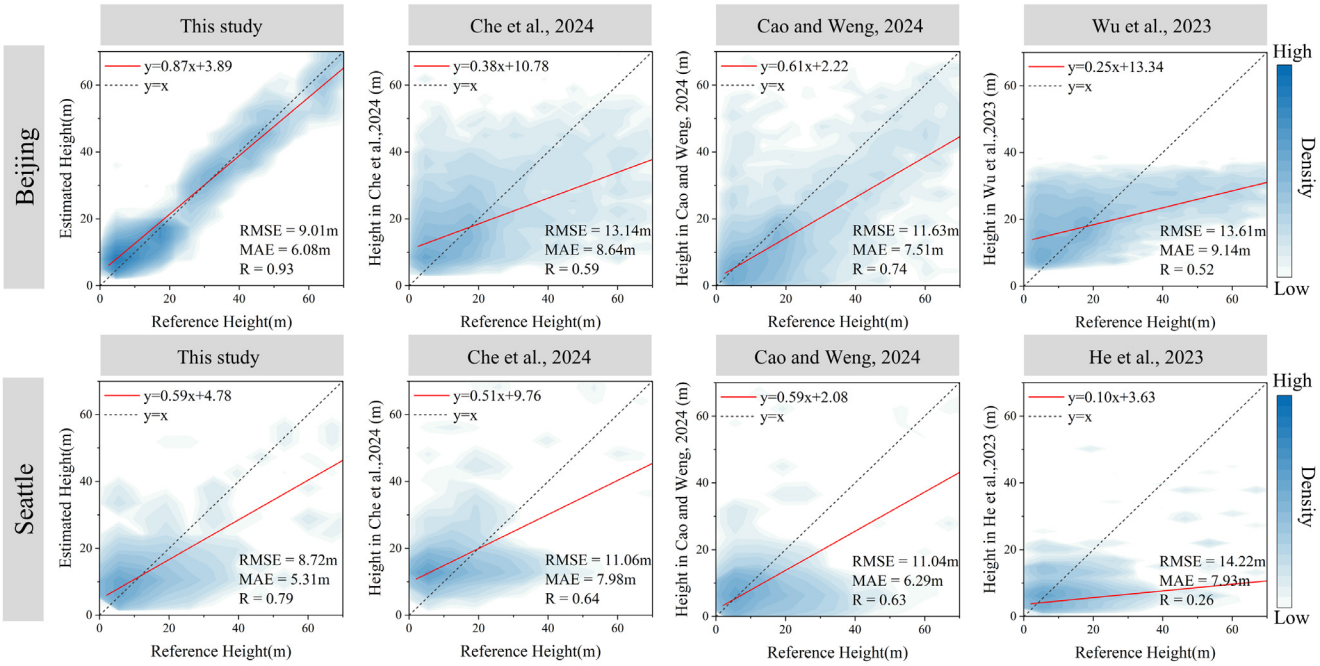


Fig. 14. Accuracy comparison between this study and existing building height products.

auxiliary validation sources to further assess the credibility of our experimental results from multiple perspectives. For Beijing, we utilized the 2019 building floor number data from Amap, converting floor counts to height estimates using the commonly accepted formula: height = number of floors $\times 3$ m. For Seattle, we employed the high-quality building height dataset released by Microsoft in 2017, which was derived through interpolation from DTM data and reported in meters. The violin plots illustrating the differences between our predicted results and both Amap's floor-converted building heights and Microsoft's DTM-interpolated heights are presented in Fig. 12. The discrepancies are tightly concentrated within a narrow range. While the error distribution in Beijing is slightly more dispersed than in Seattle, it is more tightly centered around zero, indicating higher predictive accuracy.

Furthermore, Fig. 13 shows the point density plots comparing our predicted individual building heights with the two auxiliary validation datasets, both demonstrating low estimation errors. Specifically, the RMSE and R were 10.23 m and 0.78 in Beijing and 6.59 m and 0.31 in Seattle, respectively. Despite this strong performance, we observed a tendency for low-rise buildings to be overestimated in both cities. To investigate this phenomenon, we selected representative samples and compared their predictions to corresponding street-level imagery. This qualitative validation revealed that many instances of overestimation were due to discrepancies in the reference data—such as outdated records, building reconstruction, or inaccuracies in reported floor counts—while our predictions aligned more closely with actual building heights. These findings, supported by both quantitative metrics and qualitative analysis, affirm the feasibility and effectiveness of our proposed method for large-scale, rapid estimation of individual building heights using joint spaceborne LiDAR and multisource remote sensing data.

B. Comparison With Existing Building Height Products

To further illustrate the performance of our model, we compared our results with four publicly available datasets of various resolutions. These included: 3D-GloBFP [15] (vector), height in Cao and Weng [16] (2.5-m resolution), CNBH-10 m [13] (10)-m resolution), and height in He et al. [18] (30-m resolution). The results, as illustrated in Fig. 14, clearly demonstrate the advancement of our proposed approach. Compared to 3D-GloBFP, a product that also operates at the individual-building scale, our method reduces the RMSE by 31.43% and improves the R value by 57.63% in Beijing, while achieving a 21.16% reduction in RMSE and a 23.44% improvement in R in Seattle. When compared with studies conducted at a fine grid resolution of 2.5 m, our method achieves a 22.53% decrease in RMSE and a 25.68% increase in R in Beijing, along with a 21.01% reduction in RMSE and a 25.40% improvement in R in Seattle. In comparison to two widely used products with spatial resolutions of 10 and 30 m, the advantages of our approach are even more pronounced.

Additionally, Fig. 15 presents a visual comparison of our results with other studies in both global and detailed views. First, compared to studies with the same building footprint scale, our model more clearly quantifies low-rise buildings around high-rise structures, whereas the 3D-GloBFP model tends to overestimate low-rise buildings surrounding high-rise ones. Second, for grid-based building height products, limitations in pixel resolution hinder the differentiation of various building heights, often leading to excessive diffusion of high-rise building heights into low-rise areas. Lastly, three of the grid-based products failed to adequately capture certain building heights: CNBH-10 missed buildings taller than 50 m, and the height in Cao and Weng [16] dataset failed to estimate buildings shorter than 9 m. The visual and

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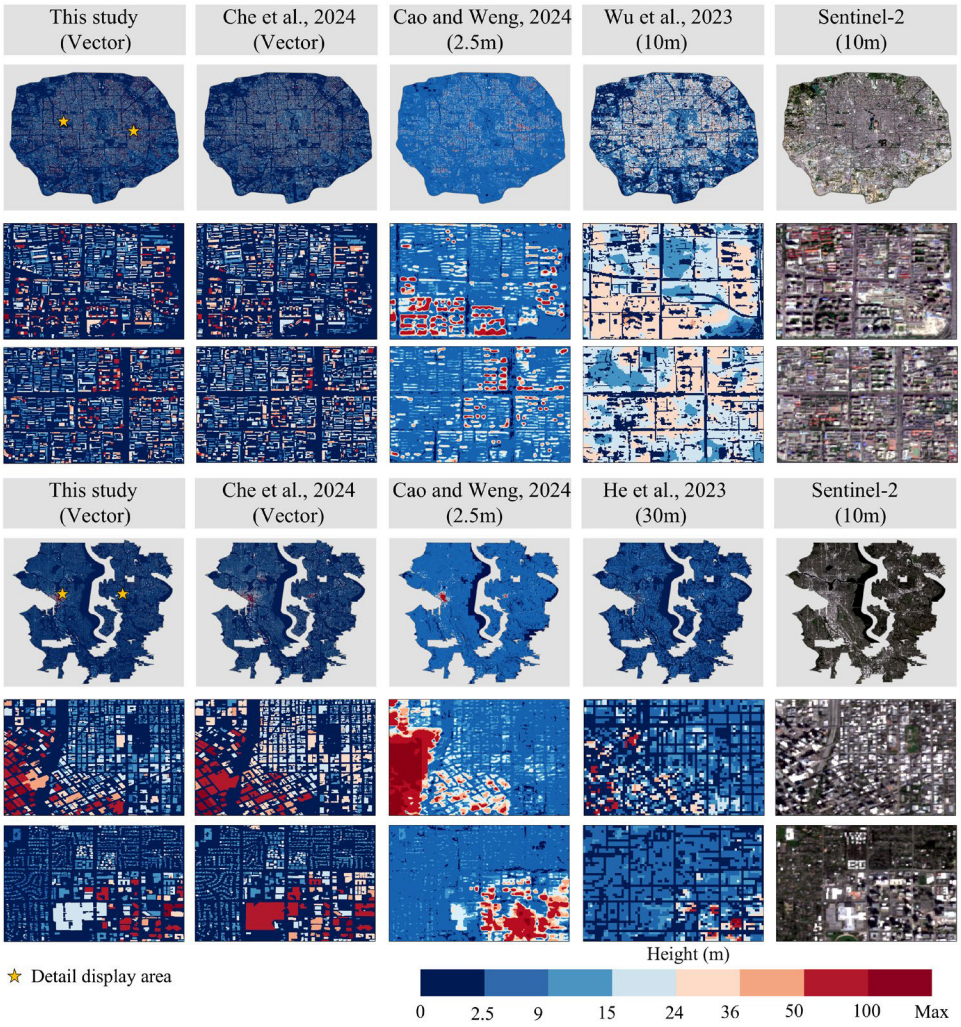


Fig. 15. Visual comparison between this study and existing building height products.

TABLE IV
SETTINGS AND ACCURACY COMPARISON OF THE BEIJING EXPERIMENT

Code	Reference	Feature	Model	RMSE	MAE	R
1	Height converted from building floor numbers	Feature set from Che et al. [15]	XGBoost	11.24	7.89	0.62
2		Features used by Che et al. [15]	NeCT-Net	7.12	4.30	0.85
3		Proposed multisource remote sensing features	NeCT-Net	6.60	4.03	0.87

quantitative results demonstrate that our method significantly improves the accuracy of building height estimation and better depicts the variation in building heights.

C. Comparison With Other Models

Since many existing large-scale building height estimation studies using Sentinel and other remote sensing imagery apply pixel-based methods, such as Random Forest and XGBoost, we first selected these two models as baselines for comparative experiments. To further assess performance beyond the limitations of classical machine learning methods, we also implemented and compared three deep learning approaches: multiscale 1-D CNN [37], LSTM [41], and Swin Transformer [42]. The comparative results are presented in Fig. 16. Compared with the widely used random forest model, our proposed method achieved substantial performance improvements: in Beijing, RMSE was reduced by 30.05%, MAE by 29.63%, and R improved by 50%; in Seattle, RMSE decreased by 10.01%, MAE by 10.46%, and R increased by 51.92%. When compared with the best-performing deep learning baseline, the Swin Transformer, our method still demonstrated clear advantages: in Beijing, RMSE decreased by 17.11%, MAE by 4.10%, and R improved by 22.23%; in Seattle, RMSE dropped by 3.21%, MAE by 3.45%, and R increased by 25.39%. To enhance the reliability of the results, we partially reproduced the method

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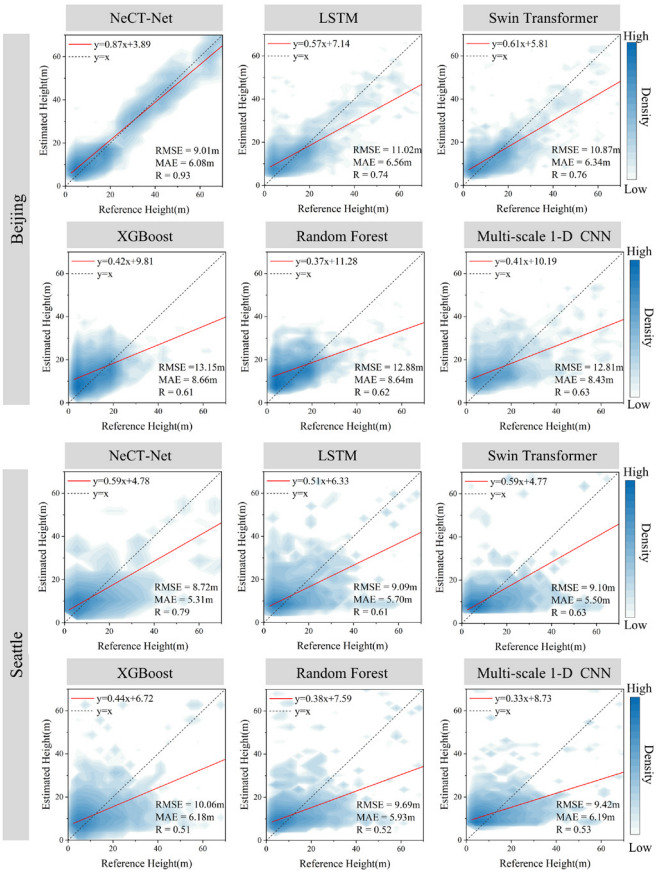


Fig. 16. Comparison of different building height estimation models.

proposed by Che et al. [15] for comparative experiments. Since the purpose of this experiment was to compare our method with Che et al. [15] in terms of features and model architecture, the subsequent semi-supervised sample enhancement and random forest-based scale conversion were not applied in this setting. The detailed experimental configuration and results are presented in Table IV. When replacing the original XGBoost model with our proposed NeCT-Net, the accuracy improved significantly. Furthermore, by substituting the original feature set with our proposed feature set, the performance was further enhanced. These results indicate that the NeCT-Net model, which takes into account the neighborhood pixel influences on building height estimation, outperforms other models and offers a novel approach for multidimensional feature integration in building height estimation.

D. Relative Importance of Variables

The weights of each layer in NeCT-Net reflect the model's capability to combine features under the overall optimization objective, rather than the direct contribution of individual input features to the final output. Therefore, it is difficult to directly quantify the specific impact of each original feature on building height estimation accuracy from the model parameters. To partially assess the contribution of various features to height estimation, we additionally employed a random forest model to evaluate feature importance. The top 30 features with the highest average relative importance in

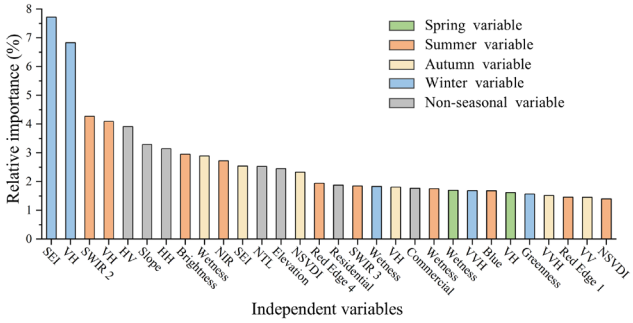


Fig. 17. Feature importance ranking based on the random forest model.

Beijing and Seattle are shown in Fig. 17. Shadow index features and polarization features demonstrated the highest contribution in the random forest model, indicating that they contain the most informative cues for capturing differences in building heights. Furthermore, features from winter and summer showed significantly higher importance than those from other seasons, suggesting that remote sensing data from these two seasons offer richer information for characterizing building height. Overall, the ranking of feature importance aligns well with our intuitive understanding of the capability of different remote sensing data sources in representing building structure.

E. Uncertainties

This study estimates individual building heights by integrating multisource remote sensing features with reference data derived from spaceborne LiDAR. Nevertheless, several sources of uncertainty remain.

1) Although the feature set was constructed using variables from diverse sensing modalities and perspectives, the selected indices may not represent the most optimal combination. As the research progresses, we plan to continually refine and supplement the feature set to include more targeted predictors for height estimation.
2) Due to the differing acquisition mechanisms of GEDI and ICESat-2 and the inherent geolocation uncertainties in both datasets, individual preprocessing and quality filtering steps are required. However, the combined use of the two datasets has already been validated in previous studies. Nonetheless, some degree of uncertainty may still remain. The parameter settings used during preprocessing largely determine the number of photons retained, and in this study, we adopted well-established parameter configurations as reported in the existing literature [43]. For broader-scale spaceborne LiDAR height retrieval tasks, future work could include ablation studies to identify more optimal preprocessing strategies.
3) Compared to directly predicting individual building heights from multisource raster data, converting pixel-level height predictions into object-level estimates using building footprint-constrained random forest regression introduces an additional source of uncertainty due to the scale transition. However, this added regression error is relatively minor in magnitude compared to the

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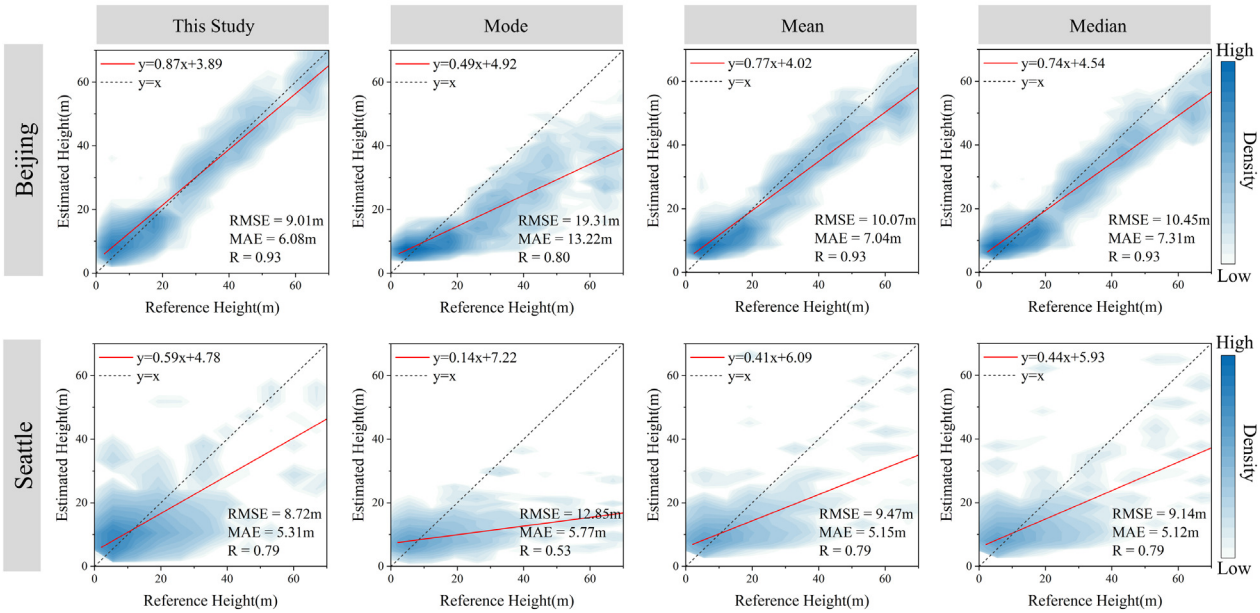


Fig. 18. Comparison of estimation accuracy during scale conversion using four methods: the proposed footprint-constrained random forest, mode, mean, and median.

uncertainties inherent in the original high-dimensional feature space. Although simple statistical methods, such as mean or median aggregation, can be used to derive building-level heights from pixel predictions, their accuracy is notably inferior to that of the random forest approach, which incorporates multiple structural indicators. As shown in Fig. 18, we compare the random forest method with three commonly used aggregation strategies—mode, mean, and median. In Beijing, the random forest model reduces the RMSE by 10.53% relative to the mean, which achieved the highest accuracy among the simple methods. In Seattle, it outperforms the best-performing median-based approach by reducing RMSE by 4.17%. These results clearly demonstrate the effectiveness of using footprint-constrained random forest regression. However, for nationwide or global applications, where model scalability is essential, using simpler aggregation methods, such as the mean or other regional statistical measures, during the scale conversion step remains an acceptable tradeoff between accuracy and computational efficiency. Our next objective is to further refine the overall research framework and extend its application to a nationwide scale, aiming to achieve accurate estimation of individual building heights across China.

VI. Conclusion

This study proposes a method for large-scale building height estimation by combining spaceborne LiDAR data with multisource remote sensing data. While most existing studies rely on building floor data converted to height, which often involves large time gaps, our method uses the latest ICESat-2 and GEDI LiDAR data to generate accurate building height samples. We incorporate a variety of spectral and polarimetric data from Sentinel-1, Sentinel-2, PALSAR, SDGSAT-1, and NASADEM to create a multimodal feature set for building height estimation. We introduce NeCT-Net that explores the relationships between multisource features and reference building height data. The method successfully estimates building heights in large urban areas, such as Beijing and Seattle, with high accuracy.

We also compare our results with the existing building height datasets and models, demonstrating that our method offers superior performance in both accuracy and clarity of building height variation. Furthermore, we incorporate semi-supervised learning techniques to address the issue of insufficient high-rise building samples, enhancing the model's performance in estimating tall buildings. Our study presents a novel and effective approach for large-scale building height estimation, providing new insights for future urban analysis.

In future work, we plan to incorporate weakly supervised learning modules into the deep learning architecture to further reduce the model's dependence on labeled samples. Moreover, since buildings with similar heights and architectural styles tend to cluster spatially, we will design a region-level height refinement module supported by road network data to further improve estimation accuracy.

References

[1] Y. Huang, T. Hong, and T. Ma, "Urban network externalities, agglomeration economies and urban economic growth," *Cities*, vol. 107, Dec. 2020, Art. no. 102882, doi: 10.1016/j.cities.2020.102882.

[2] S. Du et al., "Mapping urban functional zones with remote sensing and geospatial big data: A systematic review," *GISci. Remote Sens.*, vol. 61, no. 1, Dec. 2024, Art. no. 2404900, doi: 10.1080/15481603.2024.2404900.

[3] U. Saeed, S. R. Ahmad, G. Mohey-ud-din, H. J. Butt, and U. Ashraf, "An integrated approach for developing an urban livability composite index—A cities' ranking road map to achieve urban sustainability," *Sustainability*, vol. 14, no. 14, p. 8755, Jul. 2022, doi: 10.3390/su14148755.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 09:26:12 UTC from IEEE Xplore. Restrictions apply.

[4] C. Berger, J. Rosentreter, M. Voltersen, C. Baumgart, C. Schmullius, and S. Hese, "Spatio-temporal analysis of the relationship between 2D/3D urban site characteristics and Land Surface Temperature," *Remote Sens. Environ.*, vol. 193, pp. 225–243, May 2017, doi: 10.1016/j.rse.2017.02.020.

[5] K. Perini and A. Magliocco, "Effects of vegetation, urban density, building height, and atmospheric conditions on local temperatures and thermal comfort," *Urban Forestry Urban Greening*, vol. 13, no. 3, pp. 495–506, 2014, doi: 10.1016/j.ufug.2014.03.003.

[6] E. Resch, R. A. Bohne, T. Kvamsdal, and J. Lohne, "Impact of urban density and building height on energy use in cities," *Energy Proc.*, vol. 96, pp. 800–814, Sep. 2016, doi: 10.1016/j.egypro.2016.09.142.

[7] Y. Zhou et al., "Satellite mapping of urban built-up heights reveals extreme infrastructure gaps and inequalities in the global south," *Proc. Nat. Acad. Sci. USA*, vol. 119, no. 46, Nov. 2022, Art. no. e2214813119, doi: 10.1073/pnas.2214813119.

[8] S. Sun and C. Salvaggio, "Aerial 3D building detection and modeling from airborne LiDAR point clouds," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 3, pp. 1440–1449, Jun. 2013, doi: 10.1109/JSTARS.2013.2251457.

[9] P. Chen et al., "Leveraging Chinese GaoFen-7 imagery for high-resolution building height estimation in multiple cities," *Remote Sens. Environ.*, vol. 298, Dec. 2023, Art. no. 113802, doi: 10.1016/j.rse.2023.113802.

[10] Y. Zhao et al., "Combining ICESat-2 photons and Google Earth satellite images for building height extraction," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 117, Mar. 2023, Art. no. 103213, doi: 10.1016/j.jag.2023.103213.

[11] Y. Kaya, "Automated estimation of building heights with ICESat-2 and GEDI LiDAR altimeter and building footprints: The case of New York city and Los Angeles," *Buildings*, vol. 14, no. 11, p. 3571, Nov. 2024, doi: 10.3390/buildings14113571.

[12] D. Frantz et al., "National-scale mapping of building height using Sentinel-1 and Sentinel-2 time series," *Remote Sens. Environ.*, vol. 252, Jan. 2021, Art. no. 112128, doi: 10.1016/j.rse.2020.112128.

[13] W.-B. Wu et al., "A first Chinese building height estimate at 10 m resolution (CNBH-10 m) using multi-source Earth observations and machine learning," *Remote Sens. Environ.*, vol. 291, Jun. 2023, Art. no. 113578, doi: 10.1016/j.rse.2023.113578.

[14] M. Li, E. Koks, H. Taubenböck, and J. van Vliet, "Continental-scale mapping and analysis of 3D building structure," *Remote Sens. Environ.*, vol. 245, Aug. 2020, Art. no. 111859, doi: 10.1016/j.rse.2020.111859.

[15] Y. Che et al., "3D-GloBFP: The first global three-dimensional building footprint dataset," *Earth Syst. Sci. Data*, vol. 16, no. 11, pp. 5357–5374, Nov. 2024, doi: 10.5194/essd-16-5357-2024.

[16] Y. Cao and Q. Weng, "A deep learning-based super-resolution method for building height estimation at 2.5 m spatial resolution in the Northern Hemisphere," *Remote Sens. Environ.*, vol. 310, Aug. 2024, Art. no. 114241, doi: 10.1016/j.rse.2024.114241.

[17] X. Sun et al., "GABLE: A first fine-grained 3D building model of China on a national scale from very high resolution satellite imagery," *Remote Sens. Environ.*, vol. 305, May 2024, Art. no. 114057, doi: 10.1016/j.rse.2024.114057.

[18] T. He et al., "Global 30 meters spatiotemporal 3D urban expansion dataset from 1990 to 2010," *Scientific Data*, vol. 10, no. 1, May 2023, Art. no. 321, doi: 10.1038/s41597-023-02240-w.

[19] Y. Chen et al., "Refining urban morphology: An explainable machine learning method for estimating footprint-level building height," *Sustain. Cities Soc.*, vol. 112, Oct. 2024, Art. no. 105635, doi: 10.1016/j.scs.2024.105635.

[20] X. Yang et al., "Comprehensive LiDAR simulation with efficient physically-based DART-lux model (II): Validation with GEDI and ICESat-2 measurements at natural and urban landscapes," *Remote Sens. Environ.*, vol. 317, Feb. 2025, Art. no. 114519, doi: 10.1016/j.rse.2024.114519.

[21] X. Ma et al., "Mapping fine-scale building heights in urban agglomeration with spaceborne LiDAR," *Remote Sens. Environ.*, vol. 285, Feb. 2023, Art. no. 113392, doi: 10.1016/j.rse.2022.113392.

[22] R. Fan, H. Niu, Z. Xu, J. Chen, R. Feng, and L. Wang, "Refined urban informal settlements' mapping at agglomeration scale with the guidance of background knowledge from easy-accessed crowdsourced geospatial data," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4401716, doi: 10.1109/TGRS.2025.3527564.

[23] X. Tang et al., "A flexible framework for built-up height mapping using ICESat-2 photons and multisource satellite observations," *Remote Sens. Environ.*, vol. 318, Mar. 2025, Art. no. 114572, doi: 10.1016/j.rse.2024.114572.

[24] G. Sun et al., "Combinational shadow index for building shadow extraction in urban areas from sentinel-2A MSI imagery," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 78, pp. 53–65, Jun. 2019, doi: 10.1016/j.jag.2019.01.012.

[25] S. Du, S. Du, B. Liu, and X. Zhang, "Incorporating DeepLabv3+ and object-based image analysis for semantic segmentation of very high resolution remote sensing images," *Int. J. Digit. Earth*, vol. 14, no. 3, pp. 357–378, Mar. 2021, doi: 10.1080/17538947.2020.1831087.

[26] Q. Wang, W. Chen, Z. Huang, H. Tang, and L. Yang, "MultiSenseSeg: A cost-effective unified multimodal semantic segmentation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703724, doi: 10.1109/TGRS.2024.3390750.

[27] M. R. Pourrahmati, N. Baghdadi, and I. Fayad, "Comparison of GEDI LiDAR data capability for forest canopy height estimation over broadleaf and needleleaf forests," *Remote Sens.*, vol. 15, no. 6, p. 1522, Mar. 2023, doi: 10.3390/rs15061522.

[28] EROS Center.(2021). *GLOBAL Ecosystem Dynamics Investigation (GEDI) Level 2 User Guide*. [Online]. Available: https://lpdaac.usgs.gov/documents/591/GEDI_L2User_Guide_V2.pdf

[29] Y. Xu, S. Ding, P. Chen, H. Tang, H. Ren, and H. Huang, "Horizontal geolocation error evaluation and correction on full-waveform LiDAR footprints via waveform matching," *Remote Sens.*, vol. 15, no. 3, p. 776, Jan. 2023, doi: 10.3390/rs15030776.

[30] S. Lyu, C. Ji, Z. Liu, H. Tang, L. Zhang, and X. Yang, "Four seasonal composite Sentinel-2 images for the large-scale estimation of the number of stories in each individual building," *Remote Sens. Environ.*, vol. 303, Mar. 2024, Art. no. 114017, doi: 10.1016/j.rse.2024.114017.

[31] B. Cai, Z. Shao, X. Huang, X. Zhou, and S. Fang, "Deep learning-based building height mapping using Sentinel-1 and Sentinel-2 data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103399, doi: 10.1016/j.jag.2023.103399.

[32] S. Du, Y. Zhang, W. Sun, and B. Liu, "Quantifying heterogeneous impacts of 2D/3D built environment on carbon emissions across urban functional zones: A case study in Beijing, China," *Energy Buildings*, vol. 319, Sep. 2024, Art. no. 114513, doi: 10.1016/j.enbuild.2024.114513.

[33] Y. Cao and X. Huang, "A deep learning method for building height estimation using high-resolution multi-view imagery over urban areas: A case study of 42 Chinese cities," *Remote Sens. Environ.*, vol. 264, Oct. 2021, Art. no. 112590, doi: 10.1016/j.rse.2021.112590.

[34] P. Cai et al., "Automated building height estimation using ice, cloud, and land elevation satellite 2 light detection and ranging data and building footprints," *Remote Sens.*, vol. 16, no. 2, p. 263, Jan. 2024, doi: 10.3390/rs16020263.

[35] H. Xia et al., "Dynamic inversion method of calculating large-scale urban building height based on cooperative satellite laser altimetry and multi-source optical remote sensing," *Land*, vol. 13, no. 8, p. 1120, Jul. 2024, doi: 10.3390/land13081120.

[36] X. Li, Y. Zhou, P. Gong, K. C. Seto, and N. Clinton, "Developing a method to estimate building height from Sentinel-1 data," *Remote Sens. Environ.*, vol. 240, Apr. 2020, Art. no. 111705, doi: 10.1016/j.rse.2020.111705.

[37] S. Du, H. Liu, J. Xing, and S. Du, "Fusing multimodal data of nature-economy-society for large-scale urban building height estimation," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 129, May 2024, Art. no. 103809, doi: 10.1016/j.jag.2024.103809.

[38] Y. Jiang, J. Chang, Y. Wang, S. Wei, and D. Li, "Estimating building height using scene classification and spatial geometry," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 141, Jul. 2025, Art. no. 104675.

[39] H. Huang et al., "Estimating building height in China from ALOS AW3D30," *ISPRS J. Photogramm. Remote Sens.*, vol. 185, pp. 146–157, Mar. 2022.

[40] F. Tack, G. Buyuksalih, and R. Goossens, "3D building reconstruction based on given ground plan information and surface models extracted from spaceborne imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 67, pp. 52–64, Jan. 2012.

[41] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.

[42] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[43] P. Chen et al., "Unveiling the performance and influential factors of GEDI L2A for building height retrieval," *GISci. Remote Sens.*, vol. 62, no. 1, Dec. 2025, Art. no. 2498785.

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